FULL FAITHFULNESS OF THE FUNCTOR OF POINTS
AN INFORMAL PROOF AND ITS INTERFACES

1. Statement and size conventions

Fix a commutative ring k with carrier in $\mathcal{U}_\ell$. Test algebras are *all* commutative k-algebras with carriers in $\mathcal{U}_\ell$. Their object type belongs to $\mathcal{U}_{\ell+1}$ and their hom sets to $\mathcal{U}_\ell$. We allow functors and natural transformations on this test category to live in the larger universe. No smallness theorem for all these algebras is asserted or needed.

Schemes are qcqs locally ringed lattices with lattices and section rings at level ℓ. A k-structure is a local morphism to $\operatorname{Spec} k$; morphisms over k commute with these maps. For such a scheme X put

$$h_X(B) = \operatorname{Hom}_k(\operatorname{Spec} B, X), \qquad h_X(h)(p) = p \circ \operatorname{Spec} h \quad (h : B \to B').$$

These are sets. The identity and composition laws for Spec make h_X a covariant set-valued functor.

Theorem 1.1. *For qcqs k-schemes X, Y, postcomposition is an equivalence*

$$\operatorname{Hom}_k(X, Y) \longrightarrow \operatorname{Nat}(h_X, h_Y).$$

The proof below constructs its inverse. It does not characterise which functors are represented by schemes. In fact only the source X needs the finite affine-cover hypothesis; the descent argument works for any locally ringed lattice Y of the stated size. The construction first establishes the absolute statement on all small ring tests. Section 5 passes from that statement to tests over k.

2. Finite-cover descent for morphisms

Let X have lattice M, and let $e_1, \dots, e_n$ have join $\top$. For each i let $f_i : X|e_i \to Y$ be a local morphism. For a compact open v of Y, write $p_i(v)$ for its inverse image, viewed as an element of M below e_i. Agreement on overlaps means

$$p_i(v) \wedge e_j = p_j(v) \wedge e_i, \tag{1}$$

and agreement of the corresponding section maps after restriction to these intersections. This is equality of the restricted morphisms, expressed without identifying different presentations of a slice.

Lemma 2.1 (Separation)**.** *Two morphisms $X \to Y$ agreeing on every $X|e_i$ are equal.*

Proof. In a distributive lattice, $a = \bigvee_i (a \wedge e_i)$, so their inverse images of each v agree. Once this equality is known, their two images of a section over v lie in the same section ring. Their restrictions to the finite cover by the intersections with the e_i agree; separation of the structure sheaf gives equality. Extensionality of ringed-lattice maps, and the fact that locality is a proposition, give the result. □

Lemma 2.2 (Gluing)**.** *Compatible f_i have a unique local amalgam $f : X \to Y$.*

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Proof. Define the inverse-image function by

$$p(v) = \bigvee_i p_i(v).$$

It preserves bottom and joins, and sends top to $\bigvee_i e_i = \top$. Equation (1) and $p_i(v) \le e_i$ give

$$p(v) \wedge e_i = p_i(v).$$

Indeed each $p_j(v) \wedge e_i = p_i(v) \wedge e_j$ is below $p_i(v)$, and the ith term gives the converse inequality. Testing on the cover e_i now proves preservation of meets, since each p_i preserves meets.

For $s \in \mathcal{O}_Y(v)$, the sections $f_i^\sharp(s)$ over $p_i(v)$ agree on pairwise intersections. Glue them to a section over $p(v)$. This defines $f_v^\sharp$. Preservation of ring operations follows by separation on the same cover, because each $f_i^\sharp$ is a ring map. Naturality follows by restriction and separation on the cover of $p(w)$ when $w \le v$.

For locality, restriction of invertibility opens gives

$$D_{p(v)}(f^\sharp s) \wedge e_i = D_{p_i(v)}(f_i^\sharp s) = p_i(D_v(s)).$$

The first equality uses $D_{p(v)}(f^\sharp s) \le p(v)$, and the second is locality of f_i. These are precisely the intersections of $p(D_v(s))$ with the e_i. Lattice separation gives locality of f. The constructed morphism restricts to every f_i by its defining equations; uniqueness is Lemma 2.1. □

Restriction to an open and to a smaller open must also be identified canonically: $(X|e)|u \cong X|u$ for $u \le e$. For a morphism $g : Z \to X$, restriction to the inverse image of e gives $Z|g^*e \to X|e$. These facts follow directly from the slice universal property. All equalities of section maps can be checked after restriction in X; no transport of a whole locally ringed lattice is necessary.

3. Affine refinements

Lemma 3.1. *Every compact open of a qcqs scheme has a merely existing finite cover by affine opens. Every inverse image of a finite cover is a finite cover.*

Proof. Choose a finite affine cover e_i only while proving this mere existence. For a given compact open u, the opens $u \wedge e_i$ cover it. Through the affine chart $X|e_i \cong \operatorname{Spec} A_i$, each corresponds to an element of the Zariski lattice of A_i. Such an element merely has a finite presentation as a join of basic opens. Finite choice under truncation combines these presentations. Basic opens are affine by $(\operatorname{Spec} A_i)|D(a) \cong \operatorname{Spec}((A_i)_a)$, and the slice identifications above make these affine opens of X. Flattening the finite families gives the desired cover. The second assertion follows because inverse-image lattice maps preserve finite joins and top. □

Consequently a pair of morphisms from an open agrees whenever it agrees after every affine probe into that open: choose an affine cover and apply Lemma 2.1. The use of a mere cover is legitimate since equality of morphisms is a proposition.

4. Constructing a morphism from a natural transformation

Let $\alpha : h_X \to h_Y$ be natural. First note the following affine calculation. For an affine source $\operatorname{Spec} A$ and a natural transformation $h_{\operatorname{Spec} A} \to h_Y$, evaluation at $1_{\operatorname{Spec} A}$ produces a morphism $\operatorname{Spec} A \to Y$. Every point $q : \operatorname{Spec} B \to \operatorname{Spec} A$ over k is $\operatorname{Spec} h$ for a unique algebra map $h : A \to B$, by the universal property of Spec and $\Gamma(\operatorname{Spec} B) \cong B$. Naturality therefore determines the transformation on q by precomposition. This argument uses no small object type for the category of all test algebras.

Choose affine charts $j_i : U_i \to X$, with identifications $U_i \cong \operatorname{Spec} A_i$, and put

$$f_i = \alpha_{A_i}(j_i) : U_i \to Y.$$

For every affine probe $q : \operatorname{Spec} B \to U_i$, naturality gives

$$f_i \circ q = \alpha_B(j_i \circ q). \tag{2}$$

For a probe into $U_i \cap U_j$, the right side is the same for i and j. The overlap has a finite affine cover by Lemma 3.1; Lemma 2.1 proves that f_i and f_j agree there. Lemma 2.2 gives $f : X \to Y$ over k. To check the k-condition, compare its composite with $Y \to \operatorname{Spec} k$ to the structure map of X on the chart cover, and use separation.

For any $p : \operatorname{Spec} B \to X$, the inverse images of the U_i cover $\operatorname{Spec} B$. Refine these by finitely many basic affine opens. On such an open, with inclusion $q : \operatorname{Spec} C \to \operatorname{Spec} B$, the composite pq factors through some j_i. Equation (2) and naturality of α at $B \to C$ imply

$$f \circ p \circ q = \alpha_B(p) \circ q.$$

Separation on this cover gives $fp = \alpha_B(p)$. Thus the transformation induced by f is α. Conversely, if α is postcomposition with a given g, its chart values are gj_i, whose unique amalgam is g. These are the two inverse laws in Theorem 1.1.

The apparent choice of chart cover is harmless. The type

$$\sum_{f : \operatorname{Hom}_k(X,Y)} \prod_{B} \prod_{p : h_X(B)} (fp = \alpha_B(p))$$

is a proposition: equality of its first coordinates is detected on an affine cover, and the remaining coordinates are propositions. A cover merely exists, so it may be eliminated into this type to obtain the unique f. This also accounts for its larger universe.

5. Passing from rings to algebras over the base

Here is a precise reduction of the relative statement to the absolute one. Let $s_X : X \to \operatorname{Spec} k$ and $s_Y : Y \to \operatorname{Spec} k$ be the structure maps. For a test ring B, an ordinary point $p : \operatorname{Spec} B \to X$ determines a unique ring map $\iota : k \to B$ whose spectrum is $s_X p$. Equivalently, forgetting the algebra structure is an equivalence

$$\sum_{\iota : \operatorname{Hom}(k,B)} h^k_X(B, \iota) \simeq h_X(B).$$

For fixed p, the type of lifts on the left is a proposition, since the affine correspondence makes ι unique and the equation for being over k is a proposition. It is inhabited by that correspondence. Under a ring map $h : B \to C$, such a lift reindexes to the structure $h\iota$; uniqueness identifies it with the canonical lift of the reindexed ordinary point.

Given a transformation on k-algebras, apply it to these lifts and forget the base equations. The uniqueness just proved and relative naturality give a natural transformation on all rings. The absolute theorem recovers $f : X \to Y$. On every affine probe p, the base equation of the output says $s_Y f p = s_X p$. Affine probes detect equality, so $s_Y f = s_X$. Forgetting a relative point is injective, and every relative point is the canonical lift of its underlying ordinary point. Hence this recovered map induces the original relative transformation. The converse inverse law follows from faithfulness, again using canonical lifts of ordinary points. This proves the relative equivalence without replacing all k-algebras by finitely presented ones.

6. Proof review and implementation order

The proof requires all test algebras to be small, not finitely presented: chart rings, localisation rings, and rings of affine refinements are all allowed. Each is still at level ℓ. No chosen presentation, choice of prime ideals, or resizing principle occurs. The use of a chart cover is always under a proposition when independence matters.

The interfaces used in the proof are concrete: finite-cover separation and gluing for morphisms; nested slices and restriction of morphisms to inverse-image opens; transport of affine opens through chart isomorphisms; and the finite affine-refinement lemma. The algebraic identity and composition laws for Spec and a large-test-category functor-of-points interface then express the final inverse laws. The finite-presentation category and its Yoneda theorem are not used to bypass any of these geometric steps.

7. Connection with the implementation

The supporting geometric lemmas are now checked in `sag/qcqs/`. Finite-cover detection of opens and gluing of lattice maps are in `LatticeCover` and `LatticeDescent`. `MapAgreement` and `MapSeparation` prove separation of morphisms, with the latter phrased using actual source restrictions. `SheafMaps` glues ring maps on finite covers; `MapGluing` constructs ringed morphisms, proves their restriction equations and uniqueness, and shows that local inputs give a local amalgam. Both the ringed and local amalgam types are contractible.

`SliceMap`, `SliceNested`, and `SliceIso` provide restriction to inverse-image opens, the nested-slice isomorphism, and restriction of isomorphisms. `AffineTransport` transfers affineness. The affine case of refinement is proved in `AffineBasis`. Finite families are concatenated in `FiniteLatticeFamilies`; `AffineCoverJoin` and `AffineRefinement` extend refinement to every compact open of a qcqs scheme. These proofs include empty covers and introduce no new foundational parameters. These formed the first checkpoint, checked with 114 modules and zero failures.

The comparison is now checked, including both inverse laws. `MapLaws` supplies identity and inverse equations; `Spaces` packages the existing local maps in geometric order, without asserting univalence of the space type. `AffineTests` proves the identity and composition laws of Spec and the correspondence between affine maps and ring maps. `FunctorPoints` defines the ordinary functor of points and its natural transformations on all small rings.

`AffineEvaluation` proves evaluation for a spectrum, and `AffineSource` transfers it to any affine source. `OpenMaps`, `OpenAgreement`, and `OpenPullback` provide the inclusion triangles and overlap equations used in the proof. `TopOpen`, `AffineQcqs`, and `SliceQcqs` give the qcqs properties needed for affine probes into compact opens. `ProbeSeparation` proves that these probes detect equality. `AtlasMap` constructs the recovered map from an actual affine atlas, and `AtlasMapPoints` proves its equation on all affine tests. `PointsUniversal.Over.equivalence` eliminates the merely given atlas into the proposition of solutions and proves absolute full faithfulness.

For the relative result, `BasePoints` defines all k-algebra tests, points over k, and their reindexing laws. It also proves the equivalence between total relative points and ordinary points on a fixed test ring. `RelativeFamilies` extends a relative natural family to rings by canonical lifts, as in Section 5. `RelativeUniversal.Over.equivalence` is the statement of Theorem 1.1. Both morphism types and natural-transformation types are proved to be sets. Morphisms lie in $\mathcal{U}_\ell$ and the type of natural transformations lies in $\mathcal{U}_{\ell+1}$; the equivalence is stated at the latter level. There is no resizing or finite-presentation assumption on the tests.

The full tree checks with 133 modules and zero failures. The opening audits are recorded in `descent.opens` and `points.opens`. The remaining comparison problem is the essential-image theorem: characterising functors admitting open affine atlases and reconstructing schemes from that data. No such characterisation, or equivalence with all set-valued functors, is asserted here.